

Lineage-Aware Hierarchical Deep Learning for Imbalanced Multi-Class Classification of Peripheral Blood Cells

Progress update

Arthiga Karthigesu

3 August 2026

Where the project stands

Stage	Status
Data acquisition and verification	done
Deterministic 70/15/15 split	done
Exploratory analysis (Notebook 02)	done
Model, losses, trainer, metrics	done
Grad-CAM and significance testing	done
Backbone screen (4 backbones)	done
Main matrix, v1	invalidated
Main matrix, v2	8 of 12 runs
hierarchical $\times 3$, flat $\times 3$	done
no_imbalance	2 of 3
stain_norm	in flight
Write-up of results chapter	not started

Headline

The pipeline is complete and the proposed model trains reliably. The *scientific* claim — that lineage awareness helps rare cell types — is directionally supported but not yet statistically established.

Two things blocking the claim

- 1 Too few runs to separate the effect from training noise.
- 2 The rare classes are still the failure mode.

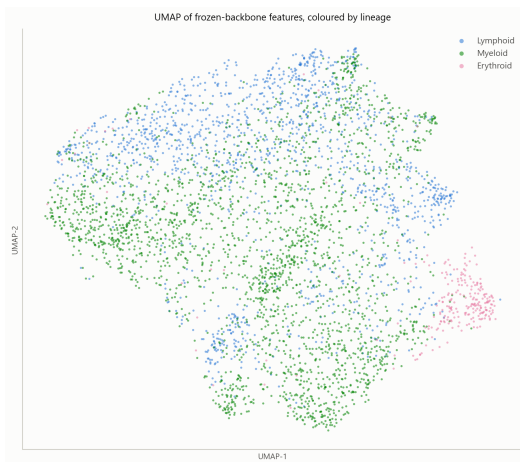
What is built — and what the data told us early

Pipeline, end to end:

- 41,621 Pappenheim-stained cells, 18 classes, 3 lineages.
- Decoded-image cache: 178 $\rightarrow$ **~ 850 img/s**. Training was data-bound, not compute-bound.
- One shared backbone $\rightarrow$ lineage head + fine head, trained jointly with a class-balanced hierarchical loss.
- One config-driven trainer runs every arm, so the ablations cannot drift apart.
- Backbone screen selected **ViT** (val macro F1 0.849).

Key early finding

Lineage structure is **local, not global**: silhouette ≈ 0 , but k -NN lineage agreement = 0.78 against a 0.45 chance baseline.



Lineages overlap heavily in frozen ImageNet features. This justifies lineage as a *soft regulariser*, and rules out a hard per-lineage routing gate.

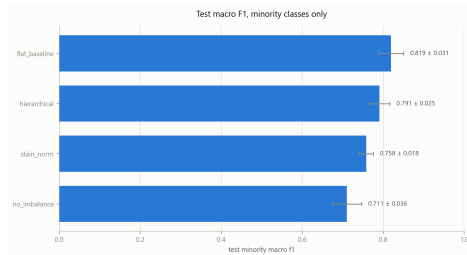
The first matrix (v1) — and why we are running it again

v1 results — ViT, budget 50 epochs, early stopping on:

Arm	Epochs survived	Macro F1	Minority F1
Hierarchical	37, 20, 29	0.853	0.791
Flat baseline	14, 13, 35	0.845	0.819
Stain-normalised	21, 14, 12	0.824	0.758
No imbalance term	13, 12, 12	0.824	0.711

The numbers looked publishable. They were not usable.

- **Run length, not method, drove the score.**
 $\text{corr}(\text{epochs run}, \text{macro F1}) = +0.87$. The arm ranking is nearly the ranking of epoch counts.
- **The learning-rate schedule was truncated.** Cosine was defined over 50 epochs but stopping fired at 12–37, leaving runs at up to **86% of peak LR** — never annealed. The 6-epoch screen runs scored *higher* than any 50-epoch run.
- **Early stopping fired on noise.** Validation macro F1 moved sd **0.021** between epochs while the effects under study were 0.008–0.028. **The measurement noise exceeded the effect.**



What v1 claimed. The flat baseline ranked *above* the hierarchical model on the rare classes — the opposite of what v2 shows. That reversal is what the confound was worth.

Why re-run

No number of extra seeds could rescue v1: the noise was larger than the effect. The protocol had to change before any comparison could mean anything.

v1 → v2 — what changed, and what it bought

Protocol changes:

Change	Why
Early stopping off , fixed 30-epoch budget	Cosine always completes; every arm gets an identical anneal
Weight EMA (0.999), evaluated instead of live weights	Halves the selection-metric noise
TTA over four flip views	Cells have no canonical orientation
RandAugment + CutMix on	Training loss had fallen to 0.005 against a 0.84 plateau — plain memorisation
<code>aug_policy</code> threaded to the dataset	It was a dead config field; v1's column was fiction

Measured on one matched config (hierarchical / ViT / seed 0 — identical data, identical seed):

Metric	v1	v2	Δ
Test macro F1	0.8695	0.8809	+0.011
Balanced accuracy	0.8797	0.8929	+0.013
Minority macro F1	0.8201	0.8370	+0.017
Cross-lineage error	0.0133	0.0119	−0.001
Within-lineage error	0.0413	0.0307	−0.011
Val F1 epoch-to-epoch sd	0.0211	0.0097	−54%

The point

The accuracy gain is modest. **Halving the measurement noise is the real win** — it is what makes an effect of 0.005–0.017 detectable at all. Cost: ~101 s/epoch versus v1's ~60 s.

Results so far — v2 matrix, ViT, 30 epochs

Arm	Seeds	Macro F1	Minority F1	Minority recall	Balanced acc.
Hierarchical (proposed)	3	0.879 $\pm$ 0.003	0.840 $\pm$ 0.007	0.799	0.894
Flat baseline	3	0.874 $\pm$ 0.006	0.823 $\pm$ 0.014	0.776	0.883
No imbalance term	2	0.870 $\pm$ 0.010	0.771 $\pm$ 0.031	0.716	0.859
Stain-normalised	0	<i>currently training</i>			

Hierarchical vs. flat (paired, by seed)

Macro F1 +0.005, $p = 0.42$, $d_z = 0.57$

Minority F1 +0.017, $p = 0.29$, $d_z = 0.84$

Reading

The proposed model wins on every metric, and by more on the rare classes than overall — which is the predicted pattern. But with $n = 3$ seeds nothing reaches significance. The effect sizes are moderate-to-large; the *sample of runs* is the limitation.

What the v2 numbers tell us

1. The imbalance term is doing the real work

Removing it costs **-0.069** minority macro F1 and **-0.083** minority recall. That is an order of magnitude more than the hierarchy contributes (+0.017).

The imbalance-robust objective is the load-bearing component of the design — which is a result worth reporting in its own right.

2. The hierarchy is nearly inert by the end of training

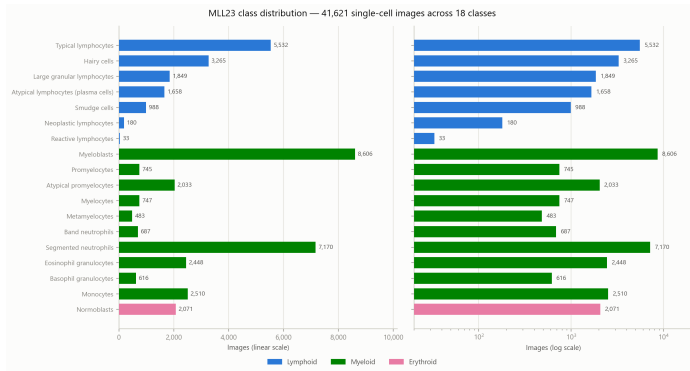
Lineage accuracy saturates at **98.9%**; both auxiliary losses collapse to ~ 0.0007 . The auxiliary task regularises an axis the model has *already solved*, so it stops contributing gradient long before training ends.

This is a *design* finding, not a tuning problem — and it points directly at the fix on the last slide.

Consistent with the exploratory analysis

Notebook 02 already found lineage separability to be local, not global, and the fine level far harder (k -NN agreement 0.78 for lineage versus 0.42 for cell type). The hard problem is discrimination *within* a lineage — myeloid maturation stages and the rare lymphoid types — which is exactly where per-class F1 still collapses.

Problem 1 — the imbalance is still the failure mode



The tail

261:1 — myeloblasts (8,606) to reactive lymphocytes (33), which leaves roughly **23 training images**. Minority recall is stuck at **0.80**.

Where F1 collapses

Reactive lymphocytes (≈ 0.55); metamyelocytes, myelocytes, band neutrophils (0.59–0.72) — the tail *and* the adjacent maturation stages.

Next — scale up the experiments

Work	Runs	Est. GPU
Finish v2 matrix (no_imbalance s2, stain_norm ×3)	4	~5 h
Raise 3 → 5 seeds on all arms	8	~10 h
Full backbone comparison at 30 epochs (ResNet / ConvNeXt / MobileNet ×3)	9	~12 h
Total	21	~27 h

Spent so far: **16.6 GPU-hours** for 8 runs on an RTX 4070 Laptop.
The 6-epoch screen *selected* a backbone; it is not the backbone *comparison* the proposal promised. Full-length runs are needed to report that honestly.

Why this matters

Three seeds gives **2 degrees of freedom**. Five seeds, combined with the halved measurement noise from EMA, is what turns the paired tests from indicative into informative.

Decision needed

27 h is feasible locally but strictly serial. A rented cloud GPU would parallelise it into a weekend.

Next — fix the imbalanced-data problem

Priority order, cheapest and highest-leverage first:

- 1 **Decoupled training (cRT).** Learn the representation with instance-balanced sampling, then retrain *only the classifier* with class-balanced sampling. Among the strongest long-tail methods, and it reuses the backbones already trained — minutes per run, not hours.
- 2 **Logit adjustment / LDAM margin loss.** Enforce larger decision margins for rare classes. A stronger lever than reweighting alone, and a drop-in replacement inside the existing `losses.py`.
- 3 **Retarget the auxiliary task.** Lineage is solved at 98.9%, so it gives no gradient late in training. Point the auxiliary head at **maturation stage within the myeloid branch**, where the residual confusion actually lives.
- 4 **Conditional inference.** Factor $p(y_2 | x) = p(y_1 | x) p(y_2 | y_1, x)$ so the coarse decision genuinely constrains the fine one, rather than two heads coexisting on a shared trunk and being nudged toward agreement.

Target

Minority macro F1 0.840 → 0.88+, and minority recall off the 0.80 plateau, with $p < 0.05$ at 5 seeds.

Sequence

Finish v2 matrix → add cRT and logit adjustment as new arms → re-run at 5 seeds → results chapter.